CHAPTER 2

THE DEVELOPMENT OF ENTERPRISE RESOURCE PLANNING SYSTEMS

LEARNING OBJECTIVES

After completing this chapter, you will be able to:

- Identify the factors that led to the development of Enterprise Resource Planning (ERP) systems
- Describe the distinguishing modular characteristics of ERP software
- Discuss the pros and cons of implementing an ERP system
- Summarize ongoing developments in ERP

INTRODUCTION

In today's competitive business environment, companies try to provide customers with goods and services faster and less expensively than their competition. How do they do that? Often, the key is an efficient, integrated information system. Increasing the efficiency of information systems results in more efficient management of business processes. When companies have efficient business processes, they can be more competitive in the marketplace.

An Enterprise Resource Planning (ERP) system can help a company integrate its operations by serving as a company-wide computing environment that includes a shared database—delivering consistent data across all business functions in real time. (The term *real time* refers to data and processes that are always current).

This chapter will help you to understand how and why ERP systems came into being and what the future might hold for business information systems. The chapter follows this sequence:

- Review of the evolution of information systems and related causes for the recent development of ERP systems
- Discussion of the few ERP software vendors that dominate the market; the current industry leader, German software maker SAP AG's industry-leading software product, SAP ERP, is discussed as an example of an ERP system.
- Review of the factors influencing a company's decision to purchase an ERP system
- Description of ERP's benefits
- Overview of frequently asked questions related to ERP systems
- Discussion of the future of ERP software, including the emphasis being placed on mobile applications and accessibility of data

THE EVOLUTION OF INFORMATION SYSTEMS

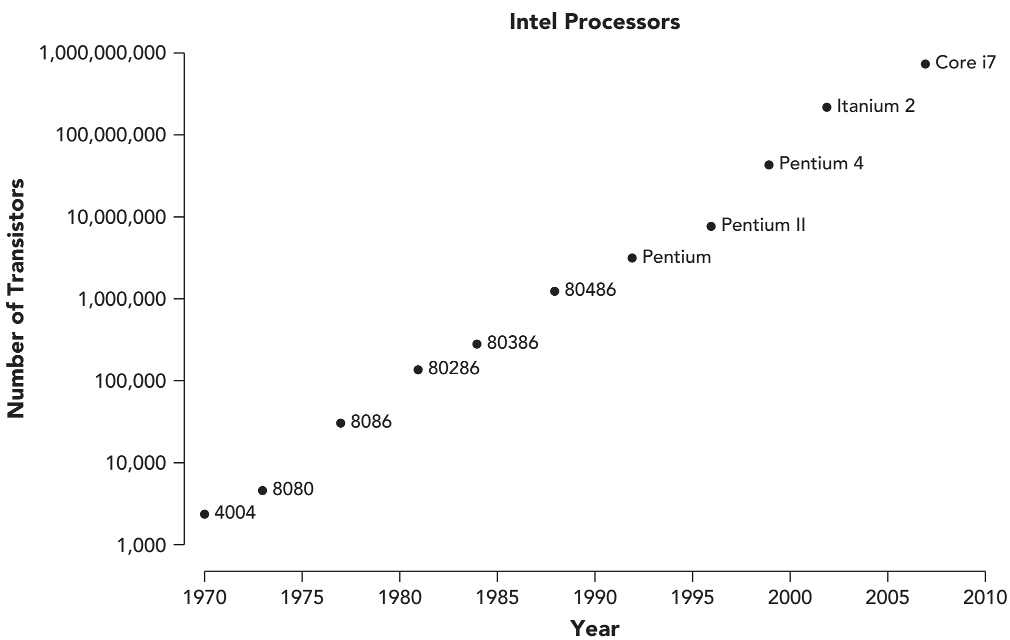
Until recently, most companies had unintegrated information systems that supported only the activities of individual business functional areas. Thus, a company would have a marketing information system, a production information system, and so on—each with its own hardware, software, and methods of processing data and information. Information systems configured in this way are known as a **silos** because each department has its own stack, or silo, of information that is unconnected to the next silo; silos are also known as stovepipes.

Such unintegrated systems might work well within each individual functional area, but to be competitive, a company must share data among all the functional areas. When a company's information systems are not integrated, costly inefficiencies can result. For example, suppose two functional areas have separate, unintegrated information systems. To share data, a clerk in one functional area needs to print out data from another area and then enter the data into her area's information system. Not only does this data input take twice the time, it also significantly increases the chance for data entry errors. Alternatively, this process might be automated by having one information system write data to a file to be read by another information system. This would reduce the probability of errors, but it could only be done periodically (usually overnight or on a weekend) to minimize the disruption to normal business transactions. Because of the time lag in updating the system, the transferred data would rarely be up to date. In addition, data can be defined differently in different data systems; for instance, products might be referred to by different part numbers in different systems. This variance can create further problems in timely and accurate information sharing between functional areas.

It seems obvious today that a business should have integrated software to manage all functional areas. An integrated ERP system, however, is an incredibly complex hardware and software system that was not feasible until the 1990s. Current ERP systems evolved as a result of three things: (1) the advancement of the hardware and software technology (computing power, memory, and communications) needed to support the system, (2) the development of a vision of integrated information systems, and (3) the reengineering of companies to shift from a functional focus to a business-process focus.

Computer Hardware and Software Development

Computer hardware and software developed rapidly in the 1960s and 1970s. The first practical business computers were the mainframe computers of the 1960s. Although these computers began to change the way business was conducted, they were not powerful enough to provide integrated, real-time data for business decision making. Over time, computers got faster, smaller, and cheaper—leading to today's proliferation of mobile devices. The rapid development of computer hardware capabilities has been accurately described by Moore's Law. In 1965, Intel employee Gordon Moore observed that the number of transistors that could be built into a computer chip doubled every 24 months. This meant that in the 1960s and 1970s the capabilities of computer hardware were doubling every 24 months, and this trend has continued, as shown in Figure 2-1.



Source Line: "The Evolution of a Revolution," ftp://download.intel.com/pressroom/kits/Intel ProcessorHistory.pdf," Intel.

FIGURE 2-1 The actual increase in transistors on a chip approximates Moore's Law

During this time, computer software was also advancing to take advantage of the increasing capabilities of computer hardware. In the 1970s, relational database software was developed, providing businesses with the ability to store, retrieve, and analyze large volumes of data. Spreadsheet software, a fundamental business tool today, became popular in the 1980s. With spreadsheets, managers could perform complex business analyses without having to rely on a computer programmer to develop custom programs.

The computer hardware and software developments of the 1960s, 1970s, and 1980s paved the way for the development of ERP systems.

Early Attempts to Share Resources

As PCs gained popularity in business in the 1980s, it became clear that users needed a way to share peripheral equipment (such as printers and hard disks, which were fairly expensive in the early 1980s) and, more importantly, data. At that point, important business information was being stored on individual PCs, but there was no easy way to share the information electronically.

By the mid-1980s, telecommunications developments allowed users to share data and peripherals on local networks. Usually, these networks were groups of computers connected to one another within a single physical location. This meant that workers could download data from a central computer to their desktop PCs and work with the data at their desks.

This central computer–local computer arrangement is now called a **client-server architecture**. Servers (central computers) became more powerful and less expensive and provided scalability. **Scalability** means that the capacity of a piece of equipment can be increased by adding new hardware. In the case of a client-server network, the ability to add servers makes the network scalable—thus extending the life of the hardware investment. Scalability is a characteristic of client-server networks, but usually not of mainframe-based systems.

By the end of the 1980s, much of the hardware and software needed to support the development of ERP systems was in place: fast computers, networked access, and advanced database technology. Recall from Chapter 1 that ERP programs help organizations manage company-wide business processes using a common database, which holds a very large amount of data. The software that holds that data in an organized fashion, and that allows for the easy retrieval of data, is the **database management system (DBMS)**. By the mid-1980s, the DBMS required to manage the development of complex ERP software existed. The final element required for the development of ERP software was understanding and acceptance from the business community. Many businesspeople did not yet recognize the benefits of integrated information systems nor were they willing to commit the resources to develop ERP software.

The Manufacturing Roots of ERP

The concept of an integrated information system took shape on the factory floor. Manufacturing software advanced during the 1960s and 1970s, evolving from simple inventory-tracking systems to **material requirements planning (MRP)** software. MRP is a production-scheduling methodology that determines the timing and quantity of production runs and purchase-order releases to meet a master production schedule. MRP software allowed a plant manager to plan production and raw materials requirements by working backward from the sales forecast, the prediction of future sales. The plant manager first looked at Marketing and Sales' forecast of customer demand, then looked at the production schedule needed to meet that demand, calculated the raw materials needed to meet the required production levels, and finally, projected the cost of those raw materials. For a company with many products, raw materials, and shared production resources, this kind of projection was impossible without a computer to keep track of various inputs.

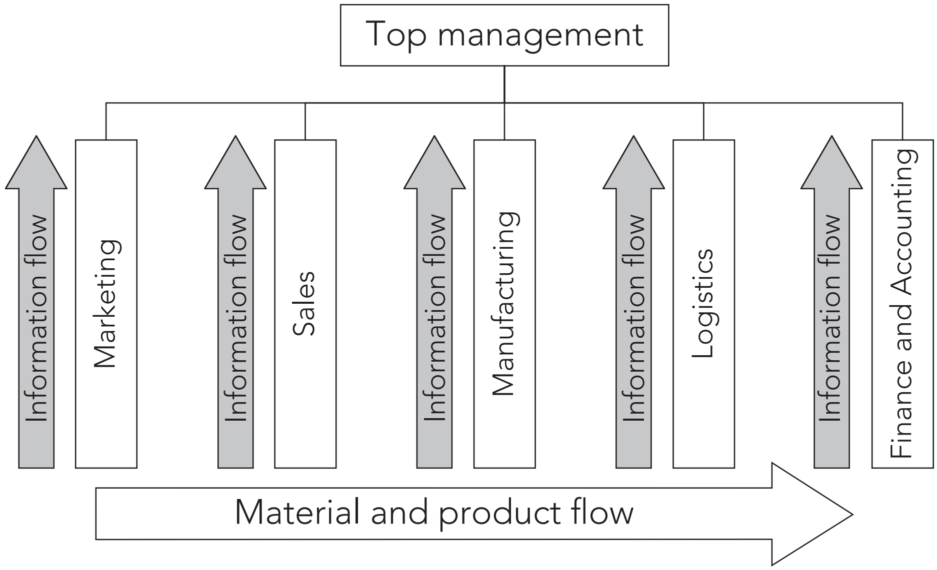
The basic functions of MRP could be handled by mainframe computers; however, the advent of **electronic data interchange (EDI)**—the direct computer-to-computer exchange of standard business documents—allowed companies to handle the purchasing process electronically, avoiding the cost and delays resulting from paper purchase order and invoice systems. The functional area now known as Supply Chain Management (SCM) began with the sharing of long-range production schedules between manufacturers and their suppliers.

Management's Impetus to Adopt ERP

The hard economic times of the late 1980s and early 1990s caused many companies to downsize and reorganize. These company overhauls were one stimulus for ERP development. Companies needed to find some way to avoid the following kind of situation (which they had tolerated for a long time):

> Imagine you are the CEO of Mountaineering, Inc., an outdoor outfitter catering to the young and trendy sportsperson. Mountaineering, Inc. is profitable and keeping pace with the competition, but the company's information systems are unintegrated and inefficient (as are the systems of your competitors). The Marketing and Sales Department creates a time-consuming and unwieldy paper trail when negotiating and closing sales with retailers. To schedule factory production, however, the manager of the Manufacturing Department needs accurate, timely information about actual and projected sales orders from the Marketing and Sales manager. Without such information, the Manufacturing manager must make a guess about which products to produce—and how many—in order to keep goods moving through the production line. Sometimes a guess overestimates demand, and sometimes it underestimates demand.
>
> Overproduction of a certain product might mean your company is stuck with a product for which there is a diminishing market due to style changes or changes in seasonal demand. When your company stores product—such as hiking boots—as it waits for a buyer, you incur warehouse expense. On the other hand, underproduction of a certain style of boot might result in product not being ready for a promised delivery date, leading to unhappy customers and, possibly, canceled orders. If you try to catch up on orders, you'll have to pay factory workers overtime or resort to the extra expense of rapid-delivery shipments.

Eventually, the management of large companies decided they could no longer afford the type of inefficiencies illustrated by the Mountaineering example—inefficiencies caused by the functional model of business organization. This model had deep roots in U.S. business, starting with the General Motors organizational model developed by Alfred P. Sloan in the 1930s. The functional business model shown in Figure 2-2 illustrates the concept of silos of information, which limit the exchange of information between the lower operating levels. Instead, the exchange of information between operating groups is handled by top management, which might not be knowledgeable about an individual functional area.

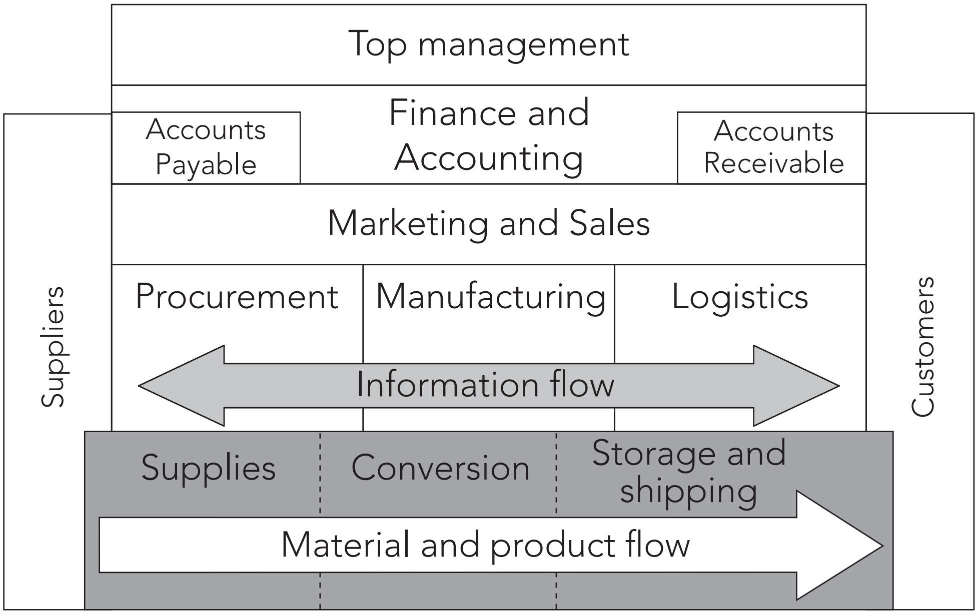


Source Line: Course Technology/Cengage Learning.

FIGURE 2-2 Information and material flows in a functional business model

The functional model was very useful for decades, and it was successful in the United States where there was limited competition and where flexibility and rapid decision making were not requirements for success. In the quickly changing markets of the 1990s, however, the functional model led to top-heavy and overstaffed organizations incapable of reacting quickly to change. The time was right to view a business as a set of cross-functional processes, as illustrated in Figure 2-3. In this organizational model, the functional business model, with its separate silos of information, is gone. Now information flows between the operating groups without top management's involvement.

In a process-oriented company, the flow of information and management activity is "horizontal" across functions, in line with the flow of materials and products. This



Source Line: Course Technology/Cengage Learning.

FIGURE 2-3 Information and material flows in a process business model

horizontal flow promotes flexibility and rapid decision making. Michael Hammer's 1993 landmark book, *Reengineering the Corporation: A Manifesto for Business Revolution,* stimulated managers to see the importance of managing business processes. Books like Hammer's, along with the difficult economic times of the late 1980s, led to a climate in which managers began to view ERP software as a solution to business problems.

In recent years, further impetus for adopting ERP systems has come from companies' efforts to be in compliance with the Sarbanes-Oxley Act of 2002, a federal law passed in response to the accounting fraud discovered at large companies such as Enron and WorldCom, among others. The law, which is covered in more detail in Chapter 5, requires companies to substantiate internal controls on all information.

In the next section, you will learn about the development of the first ERP software. SAP was the first company to develop software for ERP systems and is the current market leader in ERP software sales. As of 2011, SAP has over 109,000 customers, 53,000 employees, and sales of $12.5 billion annually.

ERP SOFTWARE EMERGES: SAP AND R/3

In 1972, five former IBM systems analysts in Mannheim, Germany—Dietmar Hopp, Claus Wellenreuther, Hasso Plattner, Klaus Tschira, and Hans-Werner Hector—formed *Systemanalyse und Programmentwicklung* (Systems Analysis and Program Development), or SAP—pronounced "S-A-P". Later, the acronym was changed to *Systeme, Anwendungen und Produkte in der Datenverarbeitung* (Systems, Applications and

Products in Data Processing.) The computer industry of the time was quite different from that of today. IBM controlled the computer market with its 360 mainframe computer, which had only 512K of main memory. In this mainframe computer environment, the SAP founders recognized that all companies developing computer software faced the same basic business problems, and each developed unique, but similar, solutions for their needs in payroll processing, accounting, materials management, and other functional areas of business. SAP's goal was to develop a standard software product that could be configured to meet the needs of each company. According to founder Dietmar Hopp, SAP's concept from the beginning was to set standards in information technology. In addition, the founders wanted data available in real time, and they wanted users to work on a computer screen rather than with voluminous printed output. These goals were lofty and forward-looking for 1972, and it took almost 20 years to achieve them.

SAP Begins Developing Software Modules

Before leaving IBM, Plattner and Hopp had worked on an order-processing system for the German chemical company ICI. The order-processing system was so successful that ICI managers also wanted a materials and logistics management system—a system for handling the purchase, receiving, and storage of materials—that could be integrated into the new order-processing system. In the course of their work for ICI, Plattner and Hopp had already developed the idea of modular software development. Software **modules** are individual programs that can be purchased, installed, and run separately, but all of the modules extract data from a common database.

In the course of their work together, Plattner and Hopp began to consider the idea of leaving IBM to form their own company so they would be free to pursue their own approach to software development. They also asked Claus Wellenreuther, an expert in financial accounting who had just left IBM, to join them, and on April 1, 1972, SAP was founded. At the time Plattner, Hopp, and Wellenreuther established the company, they could not even afford to purchase their own computer. Their first contract, with ICI, to develop the follow-on materials and logistics management system, included access to ICI's mainframe computer at night—a practice they repeated with other clients until they acquired their first computer in 1980. At ICI, the SAP founders developed their first software package, variously called System R, System RF (for real-time financial accounting), and R/1.

To keep up with the ongoing development of mainframe computer technology, in 1978 SAP began developing a more integrated version of its software products, called the R/2 system. In 1982, after four years of development, SAP released its R/2 mainframe ERP software package.

Sales grew rapidly in the 1980s, and SAP extended its software's capabilities and expanded into international markets. This was no small task, because the software had to be able to accommodate different languages, currencies, accounting practices, and tax laws.

By 1988, SAP had established subsidiaries in numerous foreign countries, launched a joint venture with consulting company Arthur Andersen, and sold its 1,000th system. SAP also became SAP AG, a publicly traded company.

SAP R/3

In 1988, SAP realized the potential of client-server hardware architecture and began development of its **R/3** system to take advantage of client-server technology. The first version of SAP R/3 was released in 1992. Each subsequent release of the SAP R/3 software contained new features and capabilities. The client-server architecture used by SAP allowed R/3 to run on a variety of computer platforms, including UNIX and Windows NT. The SAP R/3 system was also designed using an open architecture approach. In **open architecture**, third-party software companies are encouraged to develop add-on software products that can be integrated with existing software. The open architecture also makes it easy for companies to integrate their hardware products, such as bar-code scanners, personal digital assistants (PDAs), cell phones, and global information systems with the SAP system.

New Directions in ERP

In the late 1990s, the year 2000, or Y2K, problem motivated many companies to move to ERP systems. As it became clear that the date turnover from December 31, 1999, to January 1, 2000, could wreak havoc on some information systems, companies searched for ways to consolidate data, and ERP systems provided one solution.

The Y2K problem originated from programming shortcuts made by programmers in the preceding decades. With memory and storage space a small fraction of what it is today, early programmers developed software that used as few computer resources as possible. To save memory, programmers in the 1970s and 1980s typically wrote programs that only used two digits to identify a year. For example, if an invoice was posted on October 29, 1975, the programmer could just store the date as 10/29/75, rather than 10/29/1975. While this may not seem like a big storage savings, for companies with millions of transactions that needed to be stored and manipulated, it added up. These programmers never imagined that software written in the 1970s would still be running major companies and financial institutions in 1999. These old systems were known as **legacy systems**. Many companies were faced with a choice: pay programmers millions of dollars to correct the Y2K problem in their old, already outdated software—or invest in an ERP system that would not only solve the Y2K problem, but potentially provide better management of their business processes as well. Thus, the Y2K problem led to a dramatic increase in business for ERP vendors in the late 1990s. However, the rapid growth of the 1990s was followed by an ERP slump starting in 1999. By 1999, many companies were in the final stages of either an ERP implementation or modification of their existing software. Many companies that had not yet decided to move to a Y2K-compliant ERP system waited until after the new millennium to upgrade their information systems.

By 2000, SAP AG had 22,000 employees in 50 countries and 10 million users at 30,000 installations around the world. By that time, SAP also had competition in the ERP market, namely from Oracle and PeopleSoft.

PeopleSoft

PeopleSoft was founded by David Duffield, a former IBM employee who, like SAP's founders, faced opposition to his ideas from IBM management. PeopleSoft started out offering software for human resources and payroll accounting, and it achieved considerable success, even with companies that already were using SAP for accounting and production. In fact, PeopleSoft's success caused SAP to make significant modifications to its Human Resources module. In 2003, PeopleSoft strengthened its offerings in the supply chain area with its acquisition of ERP software vendor JD Edwards. Then, in late 2004, Oracle succeeded in its bid to take over PeopleSoft. Today, PeopleSoft, under

Oracle, is a popular software choice for managing human resources and financial activities at universities. Currently Oracle offers PeopleSoft's ERP solution under the PeopleSoft Enterprise Applications name; it offers JD Edwards ERP solutions as JD Edwards EnterpriseOne and JD Edwards World.

Oracle

Oracle is now SAP's biggest competitor. Oracle began in 1977 as Software Development Laboratories (SDL). Its founders, Larry Ellison, Bob Miner, and Ed Oates, won a contract from the Central Intelligence Agency (CIA) to develop a system, called Oracle, to manage large volumes of data and extract information quickly. Although the Oracle project was canceled before a successful product was developed, the three founders of SDL saw the commercial potential of a relational database system. In 1979, SDL became Relational Software, Inc., and released its first commercial database product. The company changed its name again, to Oracle, and in 1986 released the client-server Oracle relational database. The company continued to improve its database product, and in 1988 released Oracle Financials, a set of financial applications. The financial applications suite of modules included Oracle Financials, Oracle Supply Chain Management, Oracle Manufacturing, Oracle Project Systems, Oracle Human Resources, and Oracle Market Management. Oracle Financials was the foundation for what would become Oracle's ERP product.

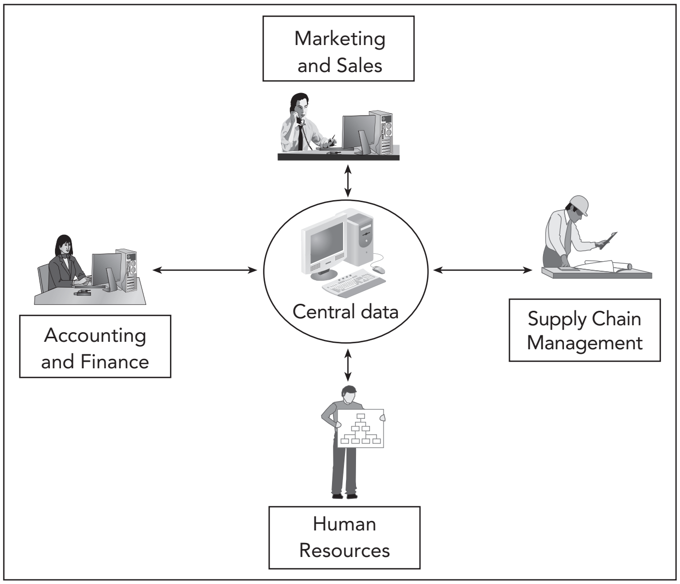
Much of Oracle's recent growth in ERP applications has been through acquisition. In addition to PeopleSoft and JD Edwards, in 2005 Oracle acquired Siebel, a major customer relationship management (CRM) software company. (CRM is discussed in more detail in Chapter 3.) In 2010, Oracle completed its acquisition of Sun Microsystems, a major manufacturer of computer hardware and software that developed the Java software development platform. According to Oracle CEO Larry Ellison, with this latest acquisition, "Oracle will be the only company that can engineer an integrated system—applications to disk—where all the pieces fit and work together so customers do not have to do it themselves. Our customers benefit as their systems integration costs go down while system performance, reliability, and security go up."

In 2010, in an effort to consolidate its customer base on a single software platform, Oracle released Fusion Applications, which is a software suite designed to give its PeopleSoft, JD Edwards, and Siebel customers a modular and flexible upgrade path to a single Oracle ERP solution.

The concepts of an Enterprise Resource Planning system are similar for the large ERP vendors, such as SAP and Oracle, and for the many smaller vendors of ERP software. Because of SAP's leadership in the ERP industry, this textbook focuses primarily on SAP's ERP software products as an example of an Enterprise Resource Planning system. Keep in mind that most other ERP software vendors provide similar functionality, with some having strengths in certain areas.

SAP ERP

SAP ERP software (previous versions were known as R/3, and later, mySAP ERP) has changed over the years, owing to product evolution—and for marketing purposes. The latest versions of ERP systems by SAP and other companies allow all business areas to access the same database, as shown in Figure 2-4, eliminating redundant data and communications lags.



Source Line: Course Technology/Cengage Learning.

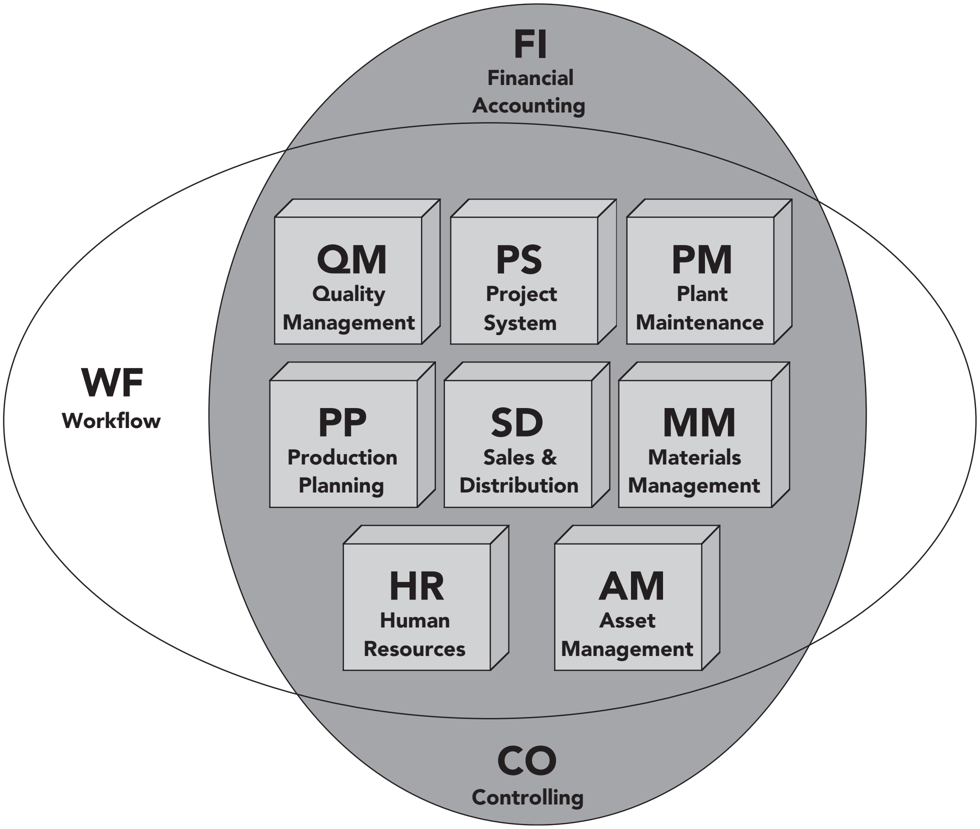
FIGURE 2-4 Data flow within an integrated information system

Perhaps most importantly, an ERP system allows data to be entered once, and then used throughout the organization. In information systems, errors most frequently occur where human beings interact with the system. ERP systems ensure that data are entered only once, where they are most likely to be accurate. For example, with access to real-time stock data, a salesperson taking an order can confirm the availability of the desired material. When the salesperson enters the sales order into the system, the order data are immediately available to Supply Chain Management, so Manufacturing can update production plans, and Materials Management can plan the delivery of the order. If the sales order data are entered correctly by the salesperson, then Supply Chain Management personnel are working with the same, correct data. The same sales data are also available to Accounting for invoice preparation.

Earlier in this chapter, you learned how software modules work. Figure 2-5 shows the major functional modules in the current SAP ERP system, also known as SAP ECC 6.0 (Enterprise Central Component 6.0), and depicts how the modules provide integration.

The basic functions of each of the modules are as follows:

- The **Sales and Distribution (SD) module** records sales orders and scheduled deliveries. Information about the customer (pricing, address and shipping instructions, billing details, and so on) is maintained and accessed from this module.
- The **Materials Management (MM) module** manages the acquisition of raw materials from suppliers (purchasing) and the subsequent handling of raw materials inventory, from storage to work-in-progress goods to shipping of finished goods to the customer.
- The **Production Planning (PP) module** maintains production information. Here production is planned and scheduled, and actual production activities are recorded.



Source Line: Course Technology/Cengage Learning.

FIGURE 2-5 Modules within the SAP ERP integrated information systems environment

- The **Quality Management (QM) module** plans and records quality control activities, such as product inspections and material certifications.
- The **Plant Maintenance (PM) module** manages maintenance resources and planning for preventive maintenance of plant machinery in order to minimize equipment breakdowns.
- The **Asset Management (AM) module** helps the company manage fixed-asset purchases (plant and machinery) and related depreciation.
- The **Human Resources (HR) module** facilitates employee recruiting, hiring, and training. This module also includes payroll and benefits.
- The **Project System (PS) module** facilitates the planning for and control over new research and development (R&D), construction, and marketing projects. This module allows for costs to be collected against a project, and it is frequently used to manage the implementation of the SAP ERP system. PS manages build-to-order items, which are low-volume, highly complex products such as ships and aircrafts.

Two financial modules, Financial Accounting (FI) and Controlling (CO), are shown in Figure 2-5 as encompassing the modules described above. That is because nearly every activity in the company has an impact on the financial position of the company:

- The **Financial Accounting (FI) module** records transactions in the general ledger accounts. This module generates financial statements for external reporting purposes.
- The **Controlling (CO) module** serves internal management purposes, assigning manufacturing costs to products and to cost centers so the profitability of the company's activities can be analyzed. The CO module supports managerial decision making.

The **Workflow (WF) module** is not a module that automates a specific business function. Rather, it is a set of tools that can be used to automate any of the activities in SAP ERP. It can perform task-flow analysis and prompt employees (by email) if they need to take action. The Workflow module works well for business processes that are not daily activities but that occur frequently enough to be worth the effort to implement the workflow module—such as preparing customer invoices.

In summary, ERP integrates business functional areas with one another. Before ERP, each functional area operated independently, using its own information systems and methods for recording transactions. ERP software also makes management reporting and decision making faster and more uniform throughout an organization. In addition, ERP promotes thinking about corporate goals, as opposed to focusing only on the goals of a single department or functional area. When top management is queried on the reasons for implementing ERP systems, the overriding answer is *control*. With the capability to see integrated data on their entire company's operation, managers use ERP systems for the control they provide, allowing managers to set corporate goals correctly.

SAP ERP Software Implementation

A truly integrated information system requires integrating all functional areas, but for various reasons, not all companies that adopt SAP software use all of the SAP ERP modules. For example, a company without factories wouldn't select the manufacturing-related modules. Another company might consider its Human Resources Department's operations to be so separate from its other operations that it would decide not integrate its Human Resources functional area. And another company might believe that its internally developed production and logistics software gives it a competitive advantage. So it might implement the SAP ERP Financial Accounting and Human Resources modules, and then integrate its internally developed production and logistics system into the SAP ERP system.

Generally, a company's level of data integration is highest when the company uses one vendor to supply all its ERP modules. When a company uses modules from different vendors, additional software must be created to get the modules to work together. Frequently, companies integrate different systems using batch data transfer processes that are performed periodically. In those cases, however, the company no longer has accurate data available in real time across the enterprise. Thus, a company must be sure the decision to use multiple vendors—or to maintain a legacy system—is based on sound business analysis, not on a resistance to change. Software upgrades of nonintegrated systems are made more problematic because further work must be done to get software from different vendors to interact. SAP's NetWeaver development platform (discussed in Chapter 8) eases the integration of SAP ERP with other software products.

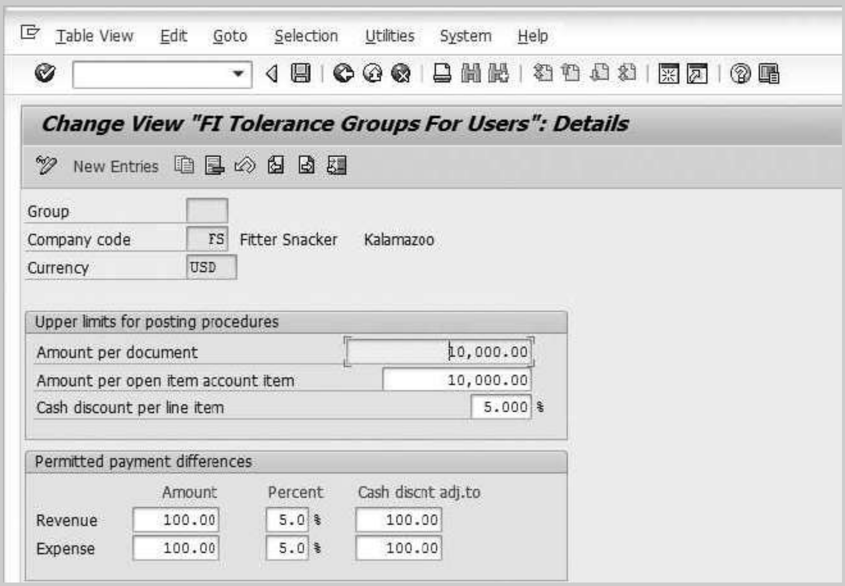
Any large software implementation is challenging—and ERP systems are no exception. There are countless examples of large implementations failing, and it is easy to understand why. Many different departments are involved, as are the many users of the system, programmers, systems analysts, and other personnel. Without top management commitment, large projects are doomed to fail. More implementation issues are discussed in Chapter 7.

After a company chooses its major modules, it must make an incredible number of decisions on how to configure the system. These configuration options allow the company to customize the modules it has chosen to fit its needs. For example, in the Financial Accounting (FI) module, a business might need to define limits on the dollar value of business transactions that certain employees can process. This is an important consideration in minimizing the risk of fraud and abuse, and is just one example of the many decisions facing a company at the start of a major ERP implementation.

ANOTHER LOOK

Tolerance Groups

In configuring the SAP system, a company can define **tolerance groups**, which are specific ranges that define transaction limits. An example of a tolerance group is shown in Figure 2-6. As part of the configuration process, a company can define any number of tolerance groups with a range of limits, and can then assign employees to these tolerance groups.



Source Line: SAP AG.

FIGURE 2-6 A customization example: tolerance groups to set transaction limits

(continued)

While SAP has defined the tolerance group methodology as its method for placing limits on an employee, configuration allows a company the flexibility to further tailor this methodology for other uses. Assume a sporting goods company places an order in January for 1,000 life jackets. The company receives only 995 in the shipment from the manufacturer, which arrives in March. This delivery, although it is short five life jackets, is close enough to the original order that it is accepted as complete. The difference of the five life jackets represents the tolerance. By defining the tolerance group to accept a variance of a small percentage of the shipment, the company has determined that it is not worth pursuing the five extra life jackets. Tolerance could indicate a shortage, as in this example, or an overabundance in an order. Thus, an order of 1,005 life jackets would also be within the tolerance. Tolerance groups should be defined and documented, in part to deal with fraud issues. The sporting goods company should know the reason for a short order: is it because the order is within the tolerance range, or is it because a worker on the loading dock stole five life jackets?

Question:

1. Can you think of other areas within a company that would need to have some limits set on variances or payments? Why would it be beneficial to set those tolerance groups?

Features of SAP ERP

Not only was SAP ERP the first software to deliver real-time ERP integration, it has other features worthy of note. Its most significant characteristics are its suitability for large companies, high cost, automation of data updates, and applicability of best practices—all of which are described below.

The original SAP ERP system targeted very large companies. Prior to the development of ERP systems, it was assumed that these giants could never have integrated systems because of the sheer amount of computing power required to integrate them. Increased computing speeds, however, meant that large companies in a variety of industries, including manufacturing, gas and oil, airlines, and consulting, could have integrated information systems.

Acquiring an SAP ERP software system is very expensive. In addition to the cost of the software, many companies find they must buy new hardware to accommodate such powerful programs. For a *Fortune* 500 company, software, hardware, and consulting costs can easily exceed $100 million. Large companies can also spend $50 million to $100 million when it is time to upgrade to a newer version of their ERP software. Full implementation of all modules can take years. In fact, most companies view ERP implementations as an ongoing process, not a one-off project. As implementations are completed in one area of a company, other areas may begin an implementation or upgrade a previous implementation.

The modular design of SAP ERP is based on business processes, such as sales order handling, materials requirement handling, and employee recruiting. When data are entered into the system, data in all related files in the central database are automatically updated. No further human input is required to make the changes.

Before the development of SAP ERP, companies often sought vendors to write software to fit their business processes. As SAP accumulated experience developing information systems, however, the company began to develop models of how certain industries' business processes should be managed. Thus, SAP ERP's design incorporates best practices for a wide variety of processes. A **best practice** is the best, most efficient way of handling a certain business process. If a company's business practices do not follow one of the best practices incorporated in the SAP ERP design, then the business must redesign its practices so it can use the software. Although some customization is possible during implementation, many companies find they must still change some of the ways they work to fit the software.

ERP FOR MIDSIZED AND SMALLER COMPANIES

By 1998, most of the *Fortune* 500 companies had already installed ERP systems, so ERP vendors refocused their marketing efforts on midsized companies (those with fewer than 1,000 employees). Small and midsized companies represented a ripe, profitable market. For example, in 2001, small and midsized companies in the United States increased their IT expenditure by more than 4 percent over the previous year. To appeal to this new market, SAP developed SAP Business All-in-One, a single package containing specific, preconfigured bundles of SAP ERP tailored for particular industries, such as automotive, banking, chemicals, and oil and gas. Because it is tailored to specific industries, SAP Business All-in-One can be installed more quickly than the standard ERP product.

To address the needs of smaller companies, such as those with less than 500 employees, SAP purchased the ERP system of Israeli-based TopManage Financial Systems in 2002, and renamed this package SAP Business One. The capabilities of SAP Business One have increased over the years through both internal development at SAP and the acquisition of other software companies such as iLytix Systems AS, which added new reporting capabilities to SAP Business One. In July 2011, SAP announced that SAP Business One would be available in a new subscription-based, hosted offering through SAP partners.

In addition, SAP has developed its BusinessByDesign product, which is an ERP product hosted by SAP and accessed through a Web browser. It is an example of the software-as-a-service (SaaS) approach that eliminates the need for a company to buy and maintain the software and hardware to run an ERP application. SaaS does not eliminate the need to configure the software to manage a company's processes or the need to train users of the system, but it does reduce the need for IT support staff to operate the ERP system. SaaS is discussed more fully in Chapter 8.

SAP and Oracle have significant competition for small-business customers from a number of smaller ERP software companies such as Sage Business Solutions, Exact, Infor, and Epicor. In 2000, software giant Microsoft acquired Great Plains, a provider of ERP software, as an entry into the small-business ERP market. Microsoft currently has five ERP products offered under the Microsoft Dynamics label.

Responses of the Software to the Changing Market

In the mid-1990s, many companies complained about the difficulty of implementing the SAP R/3 system. SAP faced a canceled implementation by Dell Computers, a lengthy implementation at Owens Corning, and a lawsuit by the now-defunct FoxMeyer drug company. Cadbury experienced a well-publicized surplus of chocolate bars at the end of 2005, due, in part, to a troubled SAP ERP implementation. In response to these and other implementation challenges, SAP developed the Accelerated SAP (ASAP) implementation methodology, a framework for implementing systems, to ease the implementation process. SAP has continued developing implementation methodology; the latest version, Solution Manager, is designed to greatly speed the implementation process.

SAP continues to extend the capabilities of SAP ERP with additional, separate products that run on separate hardware and that extract data from the SAP ERP system. In many cases, these products provide more flexible and powerful versions of tools available in the SAP ERP system. SAP's Business Warehouse (BW) product is an example of one such solution. Customers needing more capability and flexibility to analyze data beyond the standard tools and reports in the SD, PP, and FI modules can add Business Warehouse (BW). The BW software runs on a separate server and lets the user define unique reporting and analysis methods and integrate information from other systems.

The tools used to analyze data in the BW system are known as business intelligence (BI). As an example of the use of the business intelligence that can be gained through using SAP's Business Warehouse module, SAP has recently announced its in-memory data analysis tool, SAP HANA (High-performance ANalytical Appliance), a technology which will likely revolutionize Business Intelligence. Rather than storing data on hard disk drives, SAP's HANA system stores data in memory for dramatically quicker access. SAP HANA will allow companies to analyze business operations and large volumes of transactional and analytical data in real time.

The SAP ERP system provides some tools to manage customer interactions and analyze the success of promotional campaigns, but SAP also sells a separate application called Customer Relationship Management (CRM), which has extended customer service capabilities. SAP's CRM product is designed to compete with CRM systems from competitors, such as Oracle's Siebel CRM application.

SAP addressed the issue of Internet-based data exchange with its NetWeaver integration platform, which provides a unified means to connect SAP systems to other systems and to the Internet.

Like all technology, ERP software and related products are constantly changing. Thus, the challenge for a company is not only to evaluate an ERP vendor's current product offerings, but also to assess its development strategies and product plans.

CHOOSING CONSULTANTS AND VENDORS

Because ERP software packages are so large and complex, one person cannot fully understand a single ERP system; it is also impossible for an individual to adequately compare various systems. So, before choosing a software vendor, most companies study their needs and then hire an external team of software consultants to help choose the right software vendor(s) and the best approach to implementing ERP. Working as a team with the customer, the consultants apply their expertise to selecting an ERP vendor (or vendors) that will best meet their customer's needs.

After recommending a vendor, the consultants will typically recommend the software modules best suited to the company's operations, along with the configurations within those modules that are most appropriate. This preplanning should involve not only the consultants and a company's IT department, but the management of all functional business areas as well.

THE SIGNIFICANCE AND BENEFITS OF ERP SOFTWARE AND SYSTEMS

The significance of ERP lies in its many benefits. Recall that integrated information systems can lead to more efficient business processes that cost less than those in unintegrated systems. In addition, ERP systems offer the following benefits:

- ERP allows easier global integration. Barriers of currency exchange rates, language, and culture can be bridged automatically, so data can be integrated across international borders.
- ERP integrates people and data while eliminating the need to update and repair many separate computer systems. For example, at one point, Boeing had 450 data systems that fed data into its production process; the company now has a single system for recording production data.
- ERP allows management to actually manage operations, not just monitor them. For example, without ERP, getting an answer to "How are we doing?" requires getting data from each business unit and then analyzing that data for a comprehensive, integrated picture. The ERP system already has all the data, allowing the manager to focus on improving processes. This focus enhances management of the company as a whole, and makes the organization more adaptable when change is required.

An ERP system can dramatically reduce costs and improve operational efficiency. For example, EZ-FLO International, Inc., a family-run plumbing products manufacturer and distributor, had been experiencing double-digit growth and needed a scalable business platform to support its global growth. The company selected SAP Business All-in-One customized for wholesale distribution. As a result of the integrated process capabilities, EZ-FLO has been able to greatly improve its inventory management processes and has eliminated its annual inventory count, which used to take 100 employees two days to complete. In addition, the company has reduced manufacturing lead times in its domestic plants by two weeks. These process improvements led to improved customer service, and a 20 percent increase in the number of new customers—with a 12 percent growth in sales per customer.

Welch Foods, Inc., is the processing and marketing subsidiary of the National Grape Cooperative, an organization of 1,200 U.S. and Canadian grape growers. In 2004, Welch's decided to implement Oracle's E-Business Suite to make better decisions regarding product mix, production, and marketing. For Welch's, a key advantage of the ERP system is that it provides a "single source of truth" on production costs and on customer and product profitability, allowing Welch's management team to make better business decisions.

QUESTIONS ABOUT ERP

How Much Does an ERP System Cost?

The total cost of an ERP system implementation includes several factors, including the following:

- The scale of the ERP software, which corresponds to the size of the company it serves
- The need for new hardware capable of running complex ERP software
- Consultants' and analysts' fees

- Length of time required for implementation (which causes disruption of business)
- Training (which costs both time and money)

A large company, one with well over 1,000 employees, will likely spend $100 million to $500 million for an ERP system with operations involving multiple countries, currencies, languages, and tax laws. Such an installation might cost as much as $30 million in software license fees, $200 million in consulting fees, additional millions to purchase new hardware, and even more millions to train managers and employees—and full implementation of the new system could take four to six years.

A midsized company (one with fewer than 1,000 employees) might spend $10 million to $20 million in total implementation costs and have its ERP system up and running in about two years.

A smaller company, one with less than $50 million in annual revenue, could expect to pay about $300,000 for an ERP implementation, and one with revenue of $100–250 million could spend around $1.4 million. For these smaller companies, implementations usually take about 10 months.

Should Every Business Buy an ERP Package?

ERP packages imply, by their design, a certain way of doing business, and they require users to follow that way of doing business. For a particular business, some of its operations—or certain segments of its operations—might not be a good match with the constraints inherent in ERP. Therefore, it is imperative for a business to analyze its own business strategy, organization, culture, and operation *before* choosing an ERP approach.

A 1998 article in the *Harvard Business Review* provides examples that show the value of planning before trying to implement an ERP system: "Applied Materials gave up on its system when it found itself overwhelmed by the organization changes involved. Dow Chemical spent seven years and close to half a billion dollars implementing a mainframe-based enterprise system; now it has decided to start over again on a client-server version." In another example, Kmart in 2002 wrote off $130 million because of a failed ERP supply chain project. At the time, Kmart was not happy with its existing supply chain software, and it attempted to implement another product too quickly.

For years, the giant U.S. retailer Walmart chose not to purchase an ERP system, but rather to write all its software in-house. Walmart's philosophy was that the global strategic business process should drive the technology. The company's IT personnel were encouraged to consider the merchandising aspect of a process first and foremost, and then let the technology follow. However, in 2007, Walmart changed its course and decided to implement SAP Financials. The CIO of Walmart is reported to have said that implementing SAP's financial package would enable the company to grow in many countries, including China.

ERP systems are popping up in some unlikely industries. The government of Singapore is implementing an ERP project that will link 20 different healthcare providers together under one system, which is estimated to be 20 percent more efficient and will eliminate a tremendous amount of duplicate information. Along with the normal modules that an ERP system contains, such as financials and human resource management, this enterprise healthcare system will include electronic medical records and clinical management modules.

Sometimes, a company is not ready for ERP. In many cases, ERP implementation difficulties arise when management does not fully understand its current business processes and cannot make implementation decisions in a timely manner. An advantage of an ERP system is that it can reduce costs by streamlining business processes. If a company is not prepared to change its business processes to make them more efficient, then it will find itself with a large bill for software and consulting fees, with no improvement in organizational performance.

ANOTHER LOOK

Sustainability and ERP

ERP systems are helping companies track their energy consumption. SAP and Walmart are not only teaming up for the implementation of the retail giant's financial software, but they are also promoting a unified front on sustainability—focusing on reducing pollution while increasing profits—through the use of ERP software. Many SAP customers have cut energy usage; for example, 7-Eleven reduced its cooling energy usage by 12 percent, Dannon cut its fuel costs by 22 percent with better management of its transportation system, and Kendall-Jackson wine makers cut their lighting bill by 40 percent—all through the use of enhanced information from their ERP system.

Companies can use the software to track and report on their energy consumption; however, in order to make changes, a company must determine the baseline of its energy consumption. For instance, Walmart was surprised to find that only 8 percent of its carbon footprint was under the direct control of the company, the remaining 92 percent was in the company's supply chain. Since 85 percent of Walmart's suppliers run SAP, Walmart saw an opportunity to encourage energy savings on the part of the companies that supply the Walmart stores. SAP sustainability software application allows Walmart and its suppliers to track their energy usage, emissions, and consumption of other natural resources.

Question:

1. How important is sustainability to a company's bottom line? How about it its image?

Is ERP Software Inflexible?

Although many people claim that ERP systems (especially the SAP ERP system) are rigid, SAP ERP does offer numerous configuration options that help businesses customize the software to fit their unique needs. In addition, programmers can write specific routines for special applications in SAP's internal programming language, called **Advanced Business Application Programming (ABAP)**. The integration platform, NetWeaver, offers further flexibility in adding both SAP and non-SAP components to a company's IT infrastructure. Companies need to be careful about how much custom programming they include in their implementations, or they could find they have simply re-created their existing information systems in a new software package instead of gaining the benefits of improved, integrated business processes. In its implementation of PeopleSoft, FedEx Corporation installed the systems for Financial and Human Resources functions with little or no modification.

Once an ERP system is in place, trying to reconfigure it while retaining data integrity is expensive and time consuming. That is why thorough pre-implementation planning is so important. It is much easier to customize an ERP program during system configuration, before any data have been stored.

What Return Can a Company Expect from Its ERP Investment?

The financial benefits provided by an ERP system can be difficult to calculate because sometimes ERP increases revenue and decreases expenses in ways that are difficult to measure. In addition, some changes take place over such a long period of time that they are difficult to track. Finally, the old information system may not be able to provide good data on the performance of the company before the ERP implementation, making comparison difficult. Still, the return on an ERP investment can be measured and interpreted in many ways:

- Because ERP eliminates redundant effort and duplicated data, it can generate savings in operations expense. And because an ERP system can help a company produce goods and services more quickly, more sales can be generated every month.
- In some instances, a company that does not implement an ERP system might be forced out of business by competitors that have an ERP system—how do you calculate the monetary advantage of remaining in business?
- A smoothly running ERP system can save a company's personnel, suppliers, distributors, and customers much frustration—a benefit that is real, but difficult to quantify.
- Because both cost savings and increased revenue occur over many years, it is difficult to put an exact dollar figure to the amount accrued from the original ERP investment.
- Because ERP implementations take time, there may be other business factors affecting the company's costs and profitability, making it difficult to isolate the impact of the ERP system alone.
- ERP systems provide real-time data, allowing companies to improve external customer communications, which can improve customer relationships and increase sales.

How Long Does It Take to See a Return on an ERP Investment?

A **return on investment (ROI)** is an assessment of an investment project's value, calculated by dividing the value of the project's benefits by the project's costs. An ERP system's ROI can be difficult to calculate because of the many intangible costs and benefits previously mentioned. Some companies do not even try to make the calculation, on the grounds that the package is as necessary as having electricity (which is not justified as an investment project). Companies that do make the ROI calculation have seen widely varying results. Some ERP consulting firms refuse to do ERP implementations unless their client company performs an ROI. Peerstone Research reported on over 200 companies using SAP or Oracle ERP systems and found that 38 percent of survey respondents do not perform formal ROI evaluations.

In the Peerstone Research study, 63 percent of companies that did perform the calculation reported a positive ROI for ERP. Manufacturing firms are more likely to see a positive ROI than government or educational organizations. However, most companies reported that nonfinancial goals were the primary motivation for their ERP installations. Seventy-one percent of those companies surveyed said that the goal behind the ERP installation was improved management vision. Although Nestlé USA had problems with its ERP implementation, it estimated a cost savings of $325 million, after spending six years and over $200 million on the implementation.

Toro, a wholesale lawnmower manufacturer, spent $25 million and four years to implement an ERP system. At first, ROI was difficult for Toro to quantify. Then, the emergence of an expanded customer base of national retailers, such as Sears and Home Depot, made it easier to quantify benefits. With this larger pool of customers, Toro's ERP system allowed it to save $10 million in inventory costs annually—the result of better production, warehousing, and distribution methods.

A recent survey of small and midsized companies showed that only 48 percent always do an ROI evaluation, and only 25 percent always repeat the calculation after implementation. These small and midsized business owners feel they just need an ERP to support their business, even if the ROI calculation is not performed.

Why Do Some Companies Have More Success with ERP Than Others?

Early ERP implementation reports indicated that only a small percentage of companies experienced a smooth rollout of their new ERP systems *and* immediately began receiving the benefits they anticipated. However, it is important to put such reports into perspective. *All* kinds of software implementations can suffer from delays, cost overruns, and performance problems—not just ERP projects. Such delays have been a major problem for the IS industry since the early days of business computing. Nevertheless, it is worth thinking specifically about why ERP installation problems can occur.

You can find numerous cases of implementation woes in the news. W. L. Gore, the maker of GoreTex fabric, had problems implementing its PeopleSoft system for personnel, payroll, and benefits. The manufacturer sued PeopleSoft, Deloitte & Touche LLP, and Deloitte Consulting for incompetence. W. L. Gore blamed the consultants for not understanding the system and leaving its Personnel department in a mess. PeopleSoft consultants were brought in to resolve the problems after implementation, but the fix cost W. L. Gore additional hundreds of thousands of dollars.

Hershey Foods (now The Hershey Company) had a rough rollout of its ERP system in 1999, due to its use of what experts call the "Big Bang" approach to implementation, in which huge pieces of the system are implemented all at once. Companies rarely use this approach because it is so risky. Hershey's order-processing and shipping departments had glitches that were being fixed as late as September. Because of that, Hershey lost a large share of the Halloween candy market that year.

Usually, a bumpy rollout and low ROI are caused by *people* problems and misguided expectations, not computer malfunctions:

- Some executives blindly hope that new software will cure fundamental business problems that are not curable by any software. The root of a problem may lie in flawed core business processes. Unless the company changes its business processes, it will just be computerizing an ineffective way of doing business.
- Some executives and IT managers don't take enough time for a proper analysis during the planning and implementation phase.
- Some executives and IT managers skimp on employee education and training.
- Some companies do not place the ownership or accountability for the implementation project on the personnel who will operate the system. This lack of ownership can lead to a situation in which the implementation becomes an IT project rather than a company-wide project.

- Unless a large project such as an ERP installation is promoted from the top down, it is doomed to fail; top executives must be behind a project 100 percent if it is going to be successful.
- A recent academic study attempting to identify the critical success factors of ERP implementations showed that a good project manager was critical and central to success of a project. In addition, training was crucial—along with a project champion, that is, someone who might not be in the CEO role but who brings enthusiasm and leadership to a project.
- ERP implementation brings a tremendous amount of change for users of the system. Managers need to effectively manage that change in order to ensure a smooth implementation.

Many ERP implementation experts emphasize the importance of proper education and training for both employees and managers. Most people will naturally resist changing the way they do their jobs. Many analysts have noted that active top management support is crucial for successful acceptance and implementation of such company-wide changes.

Some companies willingly part with funds for software and new hardware, but they do not properly budget for employee training. ERP software is complex and can be intimidating at first. This fact alone supports the case for adequate training. Gartner Research recommends allocating 17 percent of a project's budget to training. Those companies spending less than 13 percent on training are three times more likely to have problems with their ERP implementations. The cost includes training employees on how to use the software to do their job, employees' nonproductive downtime during training, and—very importantly—educating employees about how the data they control affect the entire business operation.

Nestlé learned many lessons from its implementation of an ERP system. Its six-year, $210 million project was initially headed for failure because Nestlé didn't include on the implementation team any employees from the operating groups affected. Employees left the company, morale was down, and help desk calls were up. After three years, the ERP implementation was temporarily stopped. Nestlé USA's vice president and CIO at that time, Jeri Dunn, learned that major software implementation projects are not really about the software, rather they are about change management. "When you move to SAP, you are changing the way people work…. You are challenging their principles, their beliefs, and the way they have done things for many, many years," said Dunn. After addressing the initial problems, Nestlé ultimately reaped benefits from its ERP installation.

For many companies, it takes years before they can take full advantage of the wide variety of capabilities of their ERP systems. Most ERP installations do generate returns, and news coverage now focuses on how companies gain value from their existing systems or are upgrading and adding functionality to their existing ERP systems. Del Monte Foods needed to meet Walmart's and Target's requirements for package tracking using radio frequency identification devices (RFIDs), so approximately a year after its ERP system installation, the company tied its RFID applications into its existing SAP platform and is working to make its supply chain more efficient.

The Continuing Evolution of ERP

Understanding the social and business implications of new technologies is not easy. Howard H. Aiken, the pioneering computer engineer behind the first large-scale digital computer, the Harvard Mark I, predicted in 1947 that only six electronic digital computers would be needed to satisfy the computing needs of the entire United States. Hewlett-Packard passed up the opportunity to market the computer created by Steve Wozniak that became the Apple I. And Microsoft founder Bill Gates did not appreciate the importance of the Internet until 1995, by which time Netscape controlled the bulk of the Internet browser market. (Gates, however, did dramatically reshape Microsoft around an Internet strategy by the late 1990s. Its Internet Explorer browser is now more commonly used than any other.) Thus, even people who are most knowledgeable about a new technology do not always fully understand its capabilities or how it will change business and society.

ERP systems have been in common use only since the mid-1990s. As this young technology continues to mature, ERP vendors are working to solve the adaptability problems that plague customers. The growth in the computing power of mobile devices such as smartphones and tablets will create more opportunities for ERP companies to develop applications that provide customers with instantaneous data while improving their efficiency.

ANOTHER LOOK

ERP Allows for Bakery Chain Expansion

Au Bon Pain, the bakery and café chain based in Boston, operates 200 outlets in the United States and Asia. Over the last three years, *Health* magazine has named Au Bon Pain one of America's top five healthiest fast-food restaurant chains. The company had an interest in expanding, and Tim Oliveri, the company's chief financial officer, wanted Au Bon Pain to be able to react to market changes more rapidly while also reducing its costs. The company's existing legacy information systems were holding it back, but the company's decision to implement an SAP ERP system is now helping it achieve these goals.

The new ERP system replaced disparate systems, some of which were in paper format. This integrated enterprise system brings the storefront to the back office through a number of different module implementations. With the single system, the company is able to reduce its financial reporting cycle from weeks to days, and the system allows better management of Au Bon Pain's workforce through a Web-based portal. In addition, the new system facilitates greater compliance with financial regulations and better sales management, along with the electronic purchasing of all raw materials.

One impetus for installing this ERP system was for management to be able to digest information on market conditions, and react quickly to open new stores or renovate existing stores. For example, cafés in New York City have recently undergone major renovations, resulting in double-digit sales increases. More cafes are planned to open soon.

Question:

1. Assume you own and run a small local coffee shop. You do all your ordering of ingredients for your coffee shop by hand—using pencil, paper, mail, and telephone. All your sales are recorded by hand in a book, and transcribed for filing of small-business taxes. How could a small ERP system help your business become more efficient? What would an ERP system allow you to do?

ANOTHER LOOK

Buyer Beware: Check that Contract

The history of the ERP software industry is fraught with failed implementations. One of the latest lawsuits is by Montclair State University in New Jersey, which is suing Oracle for a failed implementation. Originally, the school contracted with Oracle (PeopleSoft) to buy an ERP package for $4.3 million, with an additional $15.75 million for the implementation. The following year, after some delays, Oracle sought an additional $8 million to complete the project. Montclair University refused the additional fee, and Oracle left the project. Now Montclair is estimating that it will take it a further $20 million to complete the implementation.

Experts agree that lawsuits such as these are not productive; no one really wins. It is essential that customers pay close inspection to the legal contract drawn up between themselves and the software vendor. ERP software is extremely complex, and it is often difficult to ask the right questions when negotiating the contracts. Most importantly, the customer should understand its own business processes and be able to relay them to the software vendor. The customer must also accept that their business processes will need to change to fit the software. To top it off, the pricing of many ERP packages is very complex, with a variety of different licensing models to choose from. The bottom line is: Understand the contract you are signing.

Question:

1. Research the topic of ERP contracts on the Internet. How could ERP vendors make it easier for customers to understand their licensing models? What could customers do to ensure a contract contains all possible eventualities?

Chapter Summary

- Several developments in business and technology allowed ERP systems to evolve to their current form:
  - The speed and power of computing hardware increased exponentially, while cost and size decreased.
  - Early client-server architecture provided the conceptual framework for multiple users sharing common data.
  - Increasingly sophisticated software facilitated integration, especially in two areas: Accounting and Finance and material requirements planning.
  - As businesses grew in size, and the business environment became more complex and competitive, business managers began to demand more efficient and competitive information systems.
  - SAP AG produced a complex, modular ERP program called R/3. The software could integrate a company's entire business by using a common database that linked all operations, allowing real-time data sharing and streamlined operations.
  - SAP R/3, now called SAP ERP, is modular software offering modules for Sales and Distribution, Materials Management, Production Planning, Quality Management, and other areas.
  - ERP software is expensive to purchase and time consuming to implement, and it requires significant employee training—but the payoffs can be spectacular. For some companies, however, the ROI may not be immediate or even calculable.
  - Experts anticipate that ERP's future focus will be on applications for mobile devices and providing instant access to large volumes of data.

Key Terms

Advanced Business Application Programming (ABAP)
Asset Management (AM) module
best practice
client-server architecture
Controlling (CO) module
database management system (DBMS)
electronic data interchange (EDI)
Financial Accounting (FI) module
Human Resources (HR) module
legacy system
material requirements planning (MRP)
Materials Management (MM) module
module
open architecture
Plant Maintenance (PM) module
Production Planning (PP) module
Project System (PS) module
Quality Management (QM) module
R/3
return on investment (ROI)
Sales and Distribution (SD) module
SAP ERP
Scalability
silo
tolerance group
Workflow (WF) module

Exercises

1. Moore's Law is said to be more of a trend, rather than a representation of the actual number of transistors on a silicon chip. What is the current status of Moore's Law? If it is not exactly holding true, what does this mean for the future of the computing industry?
2. What are the main characteristics of an ERP system? What are some newly developed features of ERP systems?
3. Imagine that your uncle owns and operates a construction company. The company owns a number of very expensive pieces of machinery, such as backhoes, for building houses and apartment buildings. Up until now, your aunt has "taken care of the books" by keeping financial records by hand. However, business is picking up, and she has gotten far behind in filing taxes, paying bills, and so on. Write a persuasive essay to your uncle about why he needs an ERP system and how it would help with not only the burden of billing, payroll, and filing taxes, but also with keeping track of the company's expensive machinery. Use the Internet for research.
4. Much has been written in the news media about ERP systems, both in print and online. Using library resources or the Internet, report on one company's positive experience with implementing an ERP system, and on another company's disappointing experience.
5. Some of examples shown in this chapter are from a traditional ERP system, SAP. Consider some smaller ERP systems. Look on the Internet at Business One by SAP, and an additional smaller system, such as Pronto software or Exact software. Compare two of the systems, and list the similarities between the module-type offerings. Are there any clear differences between them?
6. Visit *CIO* magazine's Web site, www.cio.com, and conduct a search on ERP. Based on the search results, choose an example of an ERP implementation, and write a memo to your instructor describing the implementation. Discuss ways in which you think the company adopting the ERP system could have improved its implementation.
7. [Challenge Exercise] From your university's electronic library, obtain a copy of the article, "Management Based Critical Success Factors in the Implementation of Enterprise Resource Planning Systems," by Joseph Bradley (*International Journal of Accounting Information Systems*, Volume 9, no. 3, pages 175-200, September 2008). Write a three-page paper on the findings of this study concerning factors critical to the success of an ERP implementation. Choose five factors you think are most important, and focus your writing on those five.

For Further Study and Research

Allesch, Adolf. "Thriving or Surviving? How to Take Your SAP NetWeaver Pulse." *insiderProfiles* 3, no. 3 (Summer 2007). http://insiderprofiles.wispubs.com/article.aspx?iArticleId=4672.

Au Bon Pain. "Au Bon Pain Launches Cafe Remodel Program as Part of National Expansion Strategy; Transforms New York City Cafes." *PR Newswire*, June 8, 2011. www.aubonpain.com/releases/Au%20Bon%20Pain%20Launches%20Cafe%20Remodel%20Program.pdf.

Barlas, Demir. "Wal-Mart's Confused SAP Strategy." *IT Knowledge Exchange*, May 27 2008. http://itknowledgeexchange.techtarget.com/sap-watch/wal-marts-confused-sap-strategy.

Blau, John. “SAP Arrives at Home Depot.” *InfoWorld*, June 1, 2005. www.infoworld.com/article/05/06/01/HNsaphomedepot_1.html?SUPPLY%20CHAIN%20MANAGEMENT.

Blau, John. “Whirlpool Whirls into Web Services.” *InfoWorld*, May 23, 2006. www.infoworld.com/article/06/05/23/78591_HNwhirlpoolwebservices_1.html?WEB%20SERVICES%20DEVELOPMENT.

Bradley, Joseph. “Management-Based Critical Success Factors in the Implementation of Enterprise Resource Planning Systems.” *International Journal of Accounting Information Systems* 9, no. 3 (September 2008): 175–200.

Burleson, Donald. “Four Factors that Shape the Cost of ERP.” *TechRepublic*, August 16, 2001. http://search.techrepublic.com.com/index.php?q=Four+factors+that+shape+the+cost+of+ERP&t=11&go=Search.

Business Wire. “U.S. Small and Midsize Business IT Spending Continues to Bounce Back from Recession’s Low Point, According to IDC.” June 1, 2011. www.businesswire.com/news/home/20110601005379/en/U.S.-Small-Midsize-Business-Spending-Continues-Bounce.

Chemical Week. “Connecting the Chemical Industry.” September 25, 2002.

Davenport, Thomas H. “Putting the Enterprise into the Enterprise System.” *Harvard Business Review*, (July–August 1998): 121–31.

Dignan, Larry. “Oracle Officially Launches its Fusion Apps.” *CNET*, September 20, 2010. http://news.cnet.com/8301-1001_3-20016932-92.html.

Few, Stephen. “The Information Cannot Speak for Itself.” *IntelligentEnterprise.com*, July 10, 2004. www.intelligententerprise.com/showArticle.jhtml;jsessionid=0VWVEUEXBJOASQSNDLPCKH0CJUNN2JVN?articleID=22102226.

Foroohar, Rana. “Software Savior?” *Newsweek*, January 29, 2007. www.thedailybeast.com/newsweek/2007/01/28/software-savior.html.

Forbes. “SAP Sees Orders From Mid-Sized Businesses Climbing to 45% From 30% by 2010.” April 13, 2007. www.forbes.com/afxnewslimited/feeds/afx/2007/04/13/afx3609185.html.

Greenbaum, Joshua. “The Ecosystem Advantage: It’s Not Nice to Fool Mother Nature.” *insiderProfiles* 3, no. 3 (Summer 2007). http://insiderprofiles.wispubs.com/article.aspx?iArticleId=4680.

Hammer, Michael, and James Champy. *Reengineering the Corporation*. New York: Harper Business, 1993.

Hamerman, Paul, and R. Wang. “ERP: Still a Challenge After All These Years.” *InformationWeek*, July 11, 2005. www.informationweek.com/showArticle.jhtml?articleID=165600651.

Henschen, Doug. “Wal-Mart, SAP Set Sights on Sustainability.” *InformationWeek*, January 12, 2010. www.informationweek.com/news/global-cio/roi/222301155.

Ho, Victoria. “One ERP for S’pore Non-Profit Healthcare.” *ZDNet Asia*, November 29, 2007. www.pulsesync.com/news/MediaCoverage/ZDNet%20IngoT%20Coverage_30112007.pdf.

Kanaracus, Chris. “SAP Business One Gets New Hosted, Subscription Model.” *InfoWorld*, July 5, 2011. www.infoworld.com/d/applications/sap-business-one-gets-new-hosted-subscription-model-102

Kanaracus, Chris. “SAP’s HANA In-Memory Analytics Engine Now Available.” *Computerworld*, June 20, 2011. www.computerworld.com/s/article/9217779/SAP_s_HANA_in_memory_analytics_engine_now_available.

Kirkpatrick, David. “The E-Ware War.” *Fortune*, December 7, 1998, 102–112.

Kumar, Kuldeep, and Jos van Hillegersberg. “ERP Experiences and Evolution.” *Communications of the ACM* 43, no. 4 (April 2000): 23–26.

MacDonald, Elizabeth. “W. L. Gore Alleges PeopleSoft, Deloitte Botched a Costly Software Installation.” *Wall Street Journal*, November 2, 1999.

McCue, Andy. “Too Much Candy: IT Glitch Costs Cadbury.” *BusinessWeek*, June 8, 2006. www.businessweek.com/globalbiz/content/jun2006/gb20060608_252289.htm?chan=search.

Meissner, Gerd. *SAP: Inside the Secret Software Power*. New York: McGraw-Hill, 2000.

Montgomery, Nigel. “European Retailers Divided Over ERP Versus Best of Breed.” *Tech Update*, April 18, 2003. http://techupdate.zdnet.com/techupdate/stories/main/0%2C14179%2C2913405%2C00.html.

Nadeau, Michael. “5 Keys to McKesson’s Rapid BI Transformation.” *SAP NetWeaver Magazine* 1, 2005. www.sap.com/belux/platform/netweaver/pdf/CS_NW_Magazine_McKesson.pdf.

Nelson, Emily, and Evan Ramstad. “Hershey’s Biggest Dud Is Its New Computer System.” *Wall Street Journal*, November 4, 1999.

Nolan, Sean. “By the Numbers: November 2003.” *Baseline.com*, November 1, 2003. www.baselinemag.com/article2/0,1540,1374440,00.asp.

Oracle. “Oracle Customer Snapshot, Welch Foods Inc.” 2010. www.oracle.com/us/corporate/customers/welch-foods-demantra-snapshot-150658.pdf.

Oracle. “Press Release: Oracle Completes Acquisition of Sun.” January 27, 2010. www.oracle.com/us/corporate/press/044428.

Oracle, “Press Release: Oracle Buys Sun.” April 20, 2009. www.oracle.com/us/corporate/press/018363.

Osterland, Andrew. “Blaming ERP.” *CFO.com*, January 1, 2000. www.cfo.com/article.cfm/2987370.

Peerstone Research. “ERP ROI: Myth and Reality.” *InformationWeek*, March 29, 2004. www.informationweek.com/reports/showReport.jhtml;jsessionid=5WZPDA3FLJXX0QSNDLQSKHSCJUNN2JVN?articleID=18600087&_requestid=16698.

PeopleSoft. “PeopleSoft Case Studies.” 2002. www.peoplesoft-hp.com/tools/successstories/%5Bfiles%5D/Enterprise/Automotive/Toyota%20Motor%20Manufacturing%20NA%20Drives%20Real-Time%20Productivity%20with%20PeopleSoft%208%20HRMS%20and%20Enterprise%20Portal.pdf.

Preston, Rob, “Down to Business: No One Wins an ERP Lawsuit.” *InformationWeek*, June 3, 2011. www.informationweek.com/news/global-cio/interviews/229900060?queryText=erp.

Richard, Kenneth J. “When It Comes to Enterprise Software, It’s the Contract Stupid.” *InformationWeek*, March 26, 2011. www.informationweek.com/news/global-cio/interviews/229400394?pgno=1.

SAP AG. “SAP 2011 Curriculum Congress: Run Better with SAP University Alliances.” March 11, 2011. www.sdn.sap.com/irj/uac/index?rid=/webcontent/uuid/30262b82-87d8-2d10-fa86-86ca8a9751dc.

SAP AG. “EZ-Flo International: Growing Family-Run Business Goes Global With SAP® Software.” 2008.

SAP AG. “Retail Café Chain Au Bon Pain Takes Fresh Approach with SAP.” September 19, 2006. www.sap.com/corporate-en/press.epx?pressid=6747.

SAP AG. “Welcome to the 24th Annual SAP Annual General Meeting of Shareholders.” May 25, 2011. www.sap.com/corporate-en/investors/governance/meetings/pdf/2011-05-25-ShareholderMeeting-e-presentation.pdf.

Schneider, Polly. “Human Touch Sorely Needed in ERP.” *CIO*, March 2, 1999. www.cnn.com/TECH/computing/9903/02/erpeople.ent.idg/index.html.

Slater, Derek. “How to Choose the Right ERP Software Package.” *CNN*, February 16, 1999. www.cnn.com/TECH/computing/9902/16/erppkg.ent.idg/index.html.

Sliwa, Carol. “IT Difficulties Help Take Kmart Down.” *Computerworld*, January 28, 2002. www.computerworld.com/s/article/67749/IT_difficulties_help_take_Kmart_down.

Stein, Tom. “Making ERP Add Up.” *InformationWeek*, May 24, 1999. www.informationweek.com/735/erp.htm.

Steinert-Threlkeld, Tom. “Home Depot Hopes SAP Can Help Boost Sales.” *Baseline*, May 18, 2005. www.baselinemag.com/article2/0,1397,1817341,00.asp.

Sullivan, Laurie. “ERPzilla.” *InformationWeek*, July 11, 2005. www.informationweek.com/story/showArticle.jhtml?articleID=165700832.

van Everdingen, Yvonne, Jos van Hillegersberg, and Eric Waarts. “ERP Adoption by European Midsize Companies.” *Communications of the ACM* 43, no. 4 (April 2000): 27–31.

Wailgum, Thomas. “SMB ERP Projects: Don’t Forget the ROI.” *CIO*, April 01, 2009. www.cio.com/article/487794/SMB_ERP_Projects_Don_t_Forget_the_ROI, cio.com.

Wailgum, Thomas. “University ERP: Big Mess on Campus.” *CIO*, May 1, 2005. www.cio.com/article/107706/University_ERP_Big_Mess_on_Campus.

Westervelt, Robert. “U.S. Software Sales Boost SAP earnings, Says CEO.” *SearchSAP.com*, April 22, 2004. http://searchsap.techtarget.com/originalContent/0,289142,sid21_gci961032,00.html.

Wheatley, Malcolm. “ERP Training Stinks.” *CIO*, June 1, 2000. www.cio.com/article/148900/ERP_Training_Stinks.

White, Joseph B., Don Clark, and Silvia Ascarelli. “Program of Pain.” *Wall Street Journal*, March 14, 1997.

Whiting, Rick. “At SAP, R/3 Is a Distant Memory.” *InformationWeek*, July 24, 2006. www.informationweek.com/story/showArticle.jhtml?articleID=190900725.

Whiting, Rick. “Home Depot Looks to SAP As It Modernizes.” *InformationWeek*, May 23, 2005. www.informationweek.com/story/showArticle.jhtml?articleID=163106241.

Worthen, Ben. “Extreme ERP Makeover: How to Determine If a Single-Instance ERP Implementation Is Right for You.” *CIO*, November 15, 2003. www.cio.com/article/31964/How_to_Determine_If_a_Single_Instance_ERP_Implementation_is_Right_for_You.

Worthen, Ben. “Nestlé’s Enterprise Resource Planning (ERP) Odyssey.” *CIO*, May 15, 2002. www.cio.com/article/31066/Nestle_s_Enterprise_Resource_Planning_ERP_Odyssey.